

Import-Clixml (Microsoft.PowerShell.Utility)

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Content

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Import-Clixml

Module: [Microsoft.PowerShell.Utility Module](#)

Imports a CLIXML file and creates corresponding objects in PowerShell.

Syntax

ByPath (Default)

```
Import-Clixml  
  [-Path] <String[]>  
  [-IncludeTotalCount]  
  [-Skip <UInt64>]  
  [-First <UInt64>]  
  [<CommonParameters>]
```

ByLiteralPath

```
Import-Clixml
  -LiteralPath <String[]>
  [-IncludeTotalCount]
  [-Skip <UInt64>]
  [-First <UInt64>]
  [<CommonParameters>]
```

Description

The `Import-Clixml` cmdlet imports objects that have been serialized into a Common Language Infrastructure (CLI) XML file. A valuable use of `Import-Clixml` on Windows computers is to import credentials and secure strings that were exported as secure XML using `Export-Clixml`. [Example #2]() shows how to use `Import-Clixml` to import a secure credential object.

The CLIXML data is deserialized back into PowerShell objects. However, the deserialized objects aren't a live objects. They are a snapshot of the objects at the time of serialization. The deserialized objects include properties but no methods.

The **TypeNames** property contains the original type name prefixed with `Deserialized`. [Example #3]() show the **TypeNames** property of a deserialized object.

`Import-Clixml` uses the byte-order-mark (BOM) to detect the encoding format of the file. If the file has no BOM, it assumes the encoding is UTF8.

Note

[`System.Management.Automation.ScriptBlock`] objects are serialized into the `<SKB>` element in CLIXML. However, the `<SKB>` element is always deserialized to **Strings**.

For more information about CLI, see [Language independence](#).

Examples

Example 1: Import a serialized file and recreate an object

This example uses the `Export-Clixml` cmdlet to save a serialized copy of the process information returned by `Get-Process`. `Import-Clixml` retrieves the serialized file's contents and recreates an object that is stored in the `$Processes` variable.

```
Get-Process | Export-Clixml -Path .\pi.xml
$Processes = Import-Clixml -Path .\pi.xml
```

Example 2: Import a secure credential object

In this example, given a credential that you've stored in the `$Credential` variable by running the `Get-Credential` cmdlet, you can run the `Export-Clixml` cmdlet to save the

credential to disk.

Important

`Export-Clixml` only exports encrypted credentials on Windows. On non-Windows operating systems such as macOS and Linux, credentials are exported in plain text.

```
$Credxmlpath = Join-Path (Split-Path $PROFILE) TestScript.ps1.credential
$Credential | Export-Clixml $Credxmlpath
$Credxmlpath = Join-Path (Split-Path $PROFILE) TestScript.ps1.credential
$Credential = Import-Clixml $Credxmlpath
```

The `Export-Clixml` cmdlet encrypts credential objects by using the Windows [Data Protection API](#)). The encryption ensures that only your user account can decrypt the contents of the credential object. The exported `CLIXML` file can't be used on a different computer or by a different user.

In the example, the file in which the credential is stored is represented by `TestScript.ps1.credential`. Replace **TestScript** with the name of the script with which you're loading the credential.

You send the credential object down the pipeline to `Export-Clixml`, and save it to the path, `$Credxmlpath`, that you specified in the first command.

To import the credential automatically into your script, run the final two commands. Run `Import-Clixml` to import the secured credential object into your script. This import eliminates the risk of exposing plain-text passwords in your script.

Example 3: Inspect the `TypeName` property of a deserialized object

This example shows importing an object stored as CLIXML data. The data is deserialized back into a PowerShell object. However, the deserialized object aren't a live objects. They are a snapshot of the objects at the time of serialization. The deserialized objects include properties but no methods.

```
$original = [pscustomobject] @{
    Timestamp = Get-Date
    Label     = 'Meeting event'
}
$original | Add-Member -MemberType ScriptMethod -Name GetDisplay -Value {
    '{0:yyyy-MM-dd HH:mm} {1}' -f $this.Timestamp, $this.Label
}
$original | Get-Member -MemberType ScriptMethod
```

TypeName: System.Management.Automation.PSCustomObject

Name	MemberType	Definition
----	-----	-----
Equals	Method	bool Equals(System.Object obj)
GetHashCode	Method	int GetHashCode()
GetType	Method	type GetType()
ToString	Method	string ToString()
Label	NoteProperty	string Label=Meeting event
Timestamp	NoteProperty	System.DateTime Timestamp=1/31/2024 2:27:59 PM
GetDisplay	ScriptMethod	System.Object GetDisplay();

```
$original | Export-Clixml -Path event.clixml
$deserialized = Import-CliXml -Path event.clixml
$deserialized | Get-Member
```

TypeName: Deserialized.System.Management.Automation.PSCustomObject

Name	MemberType	Definition
----	-----	-----
Equals	Method	bool Equals(System.Object obj)
GetHashCode	Method	int GetHashCode()
GetType	Method	type GetType()
ToString	Method	string ToString()
Label	NoteProperty	string Label=Meeting event
Timestamp	NoteProperty	System.DateTime Timestamp=1/31/2024 2:27:59 PM

Note that the type of the object in `$original` is **System.Management.Automation.PSCustomObject**, but the type of the object in `$deserialized` is **Deserialized.System.Management.Automation.PSCustomObject**. Also, the `GetDisplay()` method is missing from the deserialized object.

Parameters

-First

Gets only the specified number of objects. Enter the number of objects to get.

Parameter properties

| Type: | [UInt64](#) | | | Default value: | False | | | Supports wildcards: | False | | | DontShow: | False | |

Parameter sets

(All)

| Position: | Named | | | Mandatory: | False | | | Value from pipeline: | False | | | Value from pipeline by property name: | False | | | Value from remaining arguments: | False | |

-IncludeTotalCount

Reports the total number of objects in the data set followed by the selected objects. If the cmdlet can't determine the total count, it displays **Unknown total count**. The integer has an **Accuracy** property that indicates the reliability of the total count value. The value of **Accuracy** ranges from 0.0 to 1.0 where 0.0 means that the cmdlet couldn't count the objects, 1.0 means that the count is exact, and a value between 0.0 and 1.0 indicates an increasingly reliable estimate.

Parameter properties

| Type: | [SwitchParameter](#) | | | Default value: | False | | | Supports wildcards: | False | | | DontShow: | False | |

Parameter sets

(All)

| Position: | Named | | | Mandatory: | False | | | Value from pipeline: | False | | | Value from pipeline by property name: | False | | | Value from remaining arguments: | False | |

-LiteralPath

Specifies the path to the XML files. Unlike **Path**, the value of the **LiteralPath** parameter is used exactly as it's typed. No characters are interpreted as wildcards. If the path includes escape characters, enclose it in single quotation marks. Single quotation marks tell PowerShell not to interpret any characters as escape sequences.

Parameter properties

| Type: |

[String\[\]](#)

| | | Default value: | None | | | Supports wildcards: | False | | | DontShow: | False | | | Aliases: | PSPath, LP | |

Parameter sets

ByLiteralPath

| Position: | Named | | | Mandatory: | True | | | Value from pipeline: | False | | | Value from pipeline by property name: | True | | | Value from remaining arguments: | False | |

-Path

Specifies the path to the XML files.

Parameter properties

| Type: |

[String\[\]](#)

| | | Default value: | None | | | Supports wildcards: | False | | | DontShow: | False | |

Parameter sets

ByPath

| Position: | 0 | | | Mandatory: | True | | | Value from pipeline: | True | | | Value from pipeline by property name: | True | | | Value from remaining arguments: | False | |

-Skip

Ignores the specified number of objects and then gets the remaining objects. Enter the number of objects to skip.

Parameter properties

| Type: | [UInt64](#) | | | Default value: | False | | | Supports wildcards: | False | | | DontShow: | False | |

Parameter sets

(All)

| Position: | Named | | | Mandatory: | False | | | Value from pipeline: | False | | | Value from pipeline by property name: | False | | | Value from remaining arguments: | False | |

CommonParameters

This cmdlet supports the common parameters: -Debug, -ErrorAction, -ErrorVariable, -InformationAction, -InformationVariable, -OutBuffer, -OutVariable, -PipelineVariable, -ProgressAction, -Verbose, -WarningAction, and -WarningVariable. For more information, see [about_CommonParameters](#).

Inputs

String

You can pipe a string containing a path to this cmdlet.

Outputs

PSObject

This cmdlet returns objects that were deserialized from the stored XML files.

Notes

When specifying multiple values for a parameter, use commas to separate the values. For example, `<parameter-name> <value1>, <value2>`.

- [Export-Clixml](#)
- [Introducing XML Serialization](#)
- [Join-Path](#)
- [Securely Store Credentials on Disk](#)
- [Use PowerShell to Pass Credentials to Legacy Systems](#)

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